

Skills Summary

Writing and Reading Circle Equations

Use this reference while completing your worksheet or when studying.

Skill 1 — Writing the equation of a circle

Centre-radius form: $(x - h)^2 + (y - k)^2 = r^2$, centre (h, k) , radius r .

It is the distance formula with the square roots undone: the left side is the squared distance from (x, y) to the centre. So when a point on the circle is given, substitute it and read off r^2 — you never need r itself.

Example: Centre $(3, -2)$, passing through $(7, 1)$. Write the equation.

Step 1. The right side of the form is the squared distance from the centre to any point on the circle, so substituting the known point gives r^2 in one step — no square root needed.

Step 2. $r^2 = (7 - 3)^2 + (1 - (-2))^2 = 16 + 9$
 $\Rightarrow (x - 3)^2 + (y + 2)^2 = 25$

Try it: Centre $(-1, 5)$, passing through $(4, 17)$. Find r^2 .

check: 691

Skill 2 — Reading a circle equation

Reading the form: the number after the minus sign inside each bracket is the coordinate. $(x + 5)^2$ means $x - (-5)$, so $h = -5$. The right side is r^2 , never r .

Inside / on / outside: substitute the point into the left side. Less than r^2 is inside, equal is on, greater is outside.

Example: Find the radius of $(x - 6)^2 + (y + 3)^2 = 100$.

Step 1. The right-hand side of centre-radius form is the radius squared, not the radius, so the one move needed is a square root.

Step 2. $r = \sqrt{100}$
 $\Rightarrow r = 10$

Try it: Find the radius of $(x + 2)^2 + (y - 7)^2 = 144$.

check: 121

Skill 3 — Completing the square to find the centre

Method: group the x terms and the y terms, move the constant to the right, then for each group add the square of half its linear coefficient — *to both sides*. Factor each group as a perfect square.

Half of the x -coefficient becomes $-h$ inside the bracket, so a $+8x$ ends up as $(x + 4)^2$ and a centre coordinate of -4 .

Example: Put $x^2 + y^2 - 8x + 2y - 8 = 0$ into centre-radius form.

Step 1. The equation has both squared terms but also linear terms, which centre-radius form never has — that is the signal to complete the square on each variable.

Step 2. Half of -8 is -4 and half of 2 is 1 : add 16 and 1 to both sides, giving $(x - 4)^2 + (y + 1)^2 = 8 + 16 + 1$

$$\Rightarrow (x - 4)^2 + (y + 1)^2 = 25$$

Try it: Put $x^2 + y^2 + 4x - 12y + 15 = 0$ into centre-radius form and give the right-hand side.

$$(x + 2)^2 + (y - 6)^2 = 11$$

(!) Watch out: the right-hand side is r^2 . A circle written $\dots = 36$ has radius 6 , not 36 . And when you add a completing term to the left, you must add it to the right as well — forgetting that is the single most common slip in this topic.